

# SINGH NITIN RAKESH

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## Education

Uka Tarsadia University | Surat, Gujarat

B.Tech Artificial Intelligence & Data Science

Achieved a perfect **10.0 SGPA** in the final semester, demonstrating mastery of advanced coursework

Agg. CGPA: **8.68**

Sep 2021 - May 2025

## Technical Skills

**Languages** : Python(advanced), Javascript(basic), C/C++ (basic)

**Tools & Technology** : HTML, CSS, Node.js, Hugging Face, Power BI, Git, FastAPI, Flask, GitHub, Excel

**Frameworks** : Pandas, Numpy, Scikit-Learn, Matplotlib, Tensorflow, Keras, PyTorch

**Technical Domains** : Machine Learning, Deep Learning, OOP, Data Science, Computer Vision, NLP, GenAI, Manual Testing, IOT, MongoDB

**Data Analysis & Visualization** : SQL, Pandas, Numpy, Scikit-learn, Exploratory Data Analysis (EDA), Statistical Analysis, Data Cleaning

**Databases & Cloud Platforms** : Snowflake, Google Cloud Platform (GCP), AWS

## Projects

**Gender Classification API** | Node.js, FastAPI, Opencv, Caffemodel

[Github Link](#)

- This is a FastAPI app for gender detection from face images using MTCNN for face detection and a pre-trained model for classification.
- It provides one simple API endpoint that processes uploaded images and returns gender prediction results within 2-3 seconds.

**Time Series Forecasting System** | Tensorflow, Pytorch, Sarimax, Keras

[Github Link](#)

- **Developed** a forecasting system to predict future trends, providing actionable, data-driven insights for strategic planning and resource allocation.
- **Systematically benchmarked** four distinct models, including statistical (SARIMAX) and deep learning (TensorFlow, PyTorch), to identify the most accurate and reliable approach for the business problem.
- **Delivered** a final SARIMAX model with 80.90% accuracy, which outperformed more complex neural network approaches by 7.5%, demonstrating a data-backed decision to choose the most effective tool for the task.

**Face Recognition Web Application (Find You)** | RetinaFace, Facenet

[Github Link](#)

- This is a Flask-based web application for face recognition. Users can upload or capture photos to find matched faces from a dataset. The app also provides options to download all matched photos as a zip file.
- Ensuring Proper Security implementation by adding feature “Authorized access only by **Password-Protected-Link**”.
- **Achieved 96.3%** face recognition accuracy on a diverse dataset of 500+ images.

**Object Detection and Distance Measurement** | Computer Vision, YOLOv5, pytxs3

[Github Link](#)

- **Developed** a real-time object detection system using YOLOv5x model to detect multiple objects in dynamic environments with high precision.
- Implemented **custom distance measurement** algorithm.
- Integrated the system with speech output for real-time feedback, enhancing usability in practical applications like autonomous navigation.
- **Achieved 92.7%** accuracy in distance measurements within 10 meters range.

**Vaccine Uptake Prediction Model** | Numpy, Sklearn, Random Forest Algorithm

[Github Link](#)

- **Developed** a dual-prediction model to forecast individual likelihood of receiving XYZ and seasonal flu vaccines.
- **Achieved 87%** accuracy in predicting XYZ vaccine uptake, 23% above baseline.

**Solana Price Data Analysis** | Data Engineering, Data Science, Data Analysis

[Github Link](#)

- **Analyzed** over **1,300 days** of historical market data to identify key drivers of price volatility and inform potential trading strategies.
- **Engineered** a robust data pipeline to clean, process, and enrich the raw dataset with **25+ technical indicators** (e.g., RSI, MACD), creating a comprehensive foundation for analysis
- **Synthesized** the enhanced 44-column dataset, which directly led to an **8% improvement** in the predictive performance of a baseline machine learning model.

**Retail Sales Analysis on Snowflake** | Snowflake, SQL

[Github Link](#)

- **Managed** an end-to-end data analysis pipeline within **Snowflake**, processing over **1 million rows** of Rossmann retail sales data to identify key drivers for demand forecasting.
- **Executed** data cleaning, preparation, and feature engineering **entirely with SQL**, handling missing values through imputation and creating new features like IsWeekend and PromoMonthFlag.
- **Uncovered** a critical business insight through Exploratory Data Analysis (EDA): promotional periods boosted average daily sales from EUR 4,406 to EUR 7,391, a remarkable **uplift of 67.7%**

**Student Performance Analysis** | Matplotlib, Pandas, Scikit-learn

[Github Link](#)

- Analyzed a dataset of over **5,000 student records** to identify key predictors of academic success, providing actionable insights for faculty.
- Engineered an automated data pipeline using **Python for ETL** (Extract, Transform, Load) processes, ensuring a high level of data integrity (**98.5%**) for the analysis.
- Discovered a strong correlation between student engagement metrics and final grades, leading to a **data-driven recommendation** for a proactive advising program.

## Experience

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| <b>AI/ML Engineer</b>   ELaunch Solution Pvt. Ltd., Surat, Gujarat                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Dec 2024 - Present   |
| <ul style="list-style-type: none"><li>• <b>Led</b> the <b>'Solana Market Analysis'</b> project, performing end-to-end exploratory data analysis (EDA) on over <b>1,300 days</b> of historical data to identify key market drivers.</li><li>• <b>Engineered</b> a data pipeline in <b>Python</b> to clean, process, and enrich the dataset with <b>25+ technical indicators</b> (e.g., RSI, MACD), creating a comprehensive foundation for analysis.</li><li>• <b>Engineered and deployed</b> an end-to-end AI image search system, performing rigorous model evaluation and tuning, achieving <b>92% accuracy</b>.</li></ul>    |                      |
| <b>Data Analyst Intern</b>   JBcodeapp, Surat, Gujarat                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | May 2024 - June 2024 |
| <i>Project: Student Performance Analysis   Python, Numpy</i> <ul style="list-style-type: none"><li>• <b>Engineered</b> an automated data pipeline using <b>Python</b> to perform ETL (Extract, Transform, Load) on <b>5,000+ student records</b>.</li><li>• <b>Conducted</b> exploratory data analysis to identify the key factors and patterns influencing student academic performance.</li><li>• <b>Synthesized</b> findings into a comprehensive report that isolated key factors influencing academic scores, providing a foundation for <b>data-driven</b> teaching methods and resource allocation strategies.</li></ul> |                      |
| <b>AI/ML Trainee</b>   AlgoBrain AI, Surat, Gujarat                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sep 2023 - Jan 2024  |
| <i>Project: Movie-Ticket Booking Chatbot with DialogFlow CX</i> <ul style="list-style-type: none"><li>• <b>Analyzed</b> potential user conversation paths to design and build a movie-ticket booking chatbot in DialogFlow CX, enhancing the overall user experience.</li><li>• <b>Systematically investigated</b> 4 web applications to identify and resolve <b>30+ critical bugs</b>, showcasing strong analytical skills in <b>anomaly detection</b> and root cause analysis.</li></ul>                                                                                                                                      |                      |

## Extracurricular Activities

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| <b>Research Community Member</b>   Swaayan Drone Club, SVNIT, Surat, Gujarat                                                                                       | May 2024 - June 2024 |
| <ul style="list-style-type: none"><li>• Built and Assisted for Drone Bootcamps by building and calibrating drones using beta flight and mission planner.</li></ul> |                      |
| <b>User Software Tester (Part-Time)</b>   Nothing India, Remote                                                                                                    | Feb 2023 - Present   |
| <ul style="list-style-type: none"><li>• Conducted user trial testing for Nothing Phone 1, Nothing Phone 2, Nothing 2a, Android 14, Cmf Watch, and more.</li></ul>  |                      |
| <b>Summer Analytics</b>   IIT Guwahati, Online                                                                                                                     | July 2024 - Aug 2024 |
| <ul style="list-style-type: none"><li>• Successfully completed Summer Analytics 2024 Program with top 10 ranking out of all participants.</li></ul>                |                      |
| <b>Aspire Leaders Program</b>   Aspire Institute, Harvard Business School, Online                                                                                  | Aug 2024 - Nov 2024  |
| <ul style="list-style-type: none"><li>• Completed all modules in Leadership, Communication, and Problem-Solving.</li></ul>                                         |                      |
| <b>AI Business Fellow</b>   Perplexity, Online                                                                                                                     | Feb 2025 - Aug 2025  |
| <ul style="list-style-type: none"><li>• Selected in the Perplexity AI Business Fellow Program 2025.</li></ul>                                                      |                      |
| <b>Gujarat State Lead</b>   Open Source Connect, Remote                                                                                                            | Jul 2025 – Sep 2025  |
| <ul style="list-style-type: none"><li>• Led Open Source Connect initiatives in Gujarat, driving AI, ML, and marketing outreach efforts.</li></ul>                  |                      |
| <b>McKinsey.org Forward Program</b>   McKinsey.org, Online                                                                                                         | Apr 2025 – Jun 2025  |
| <ul style="list-style-type: none"><li>• Completed professional development in business analytics and strategic thinking</li></ul>                                  |                      |

## Achievements

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| <ul style="list-style-type: none"><li>• Awarded a <b>Certificate of Excellence</b> in Summer Analytics (IIT Guwahati) for finishing in the top 10 of all participants.</li><li>• Certified in <b>Time Series Analysis</b> by <b>IIT Guwahati</b>, with demonstrated expertise in statistical forecasting models.</li><li>• Achieved <b>Runner-Up</b> position in Tattva Hackathon 2025 (CGPIT), demonstrating problem-solving, teamwork, and innovation.</li><li>• Achieved <b>Winner</b> position in Quiz (Group) Competition at College Fest 2023, showcasing knowledge, teamwork, and quick thinking.</li><li>• Secured <b>2nd Runner-Up</b> position in Script Writing (Solo) at College Fest 2023, demonstrating creativity and communication skills.</li><li>• Selected to participate in <b>Finnovate and Amalthea Hackathons (IITGN)</b>, reflecting innovation and technical proficiency.</li></ul> |  |
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## Certifications

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| <ul style="list-style-type: none"><li>• Passed Google Analytics Certification. <a href="#">[View Credential]</a></li><li>• Python for Data Science by NPTEL. <a href="#">[View Credential]</a></li><li>• BOOT CAMP 8.0 on Drone Technology and Applications by SVNIT. <a href="#">[View Credential]</a></li><li>• Recent Advancements in AI and Robotics by SVNIT. <a href="#">[View Credential]</a></li><li>• Completed AI/ML for GeoData Analysis by ISRO. <a href="#">[View Credential]</a></li><li>• Completed ML: An Overview from Coursera. <a href="#">[View Credential]</a></li><li>• Achieved Postman API Fundamentals Student Expert Certification, Postman. <a href="#">[View Credential]</a></li></ul> |  |
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